Topic B: common theory and application costs

Human Compression project

The common paper and four applications recover thirty declared clauses from manuscripts by Buekenhout–Cara (1998), Cameron–Cara (2002), Archer–Cara–Krempa (2005), and Garcia-Colin–Leemans (the fixed revised manuscript associated with the 2021 publication). Their retained inputs, source corrections and exclusions are unchanged by this measurement.

One complete topic and four conditional comparisons

The common theory costs **14,119** characters. The four applications cost **44,942**, giving the complete written denominator **59,061**. These figures use NFC characters without whitespace in complete extracted PDF text, after the exact corrections and furniture exclusions described on the next page.

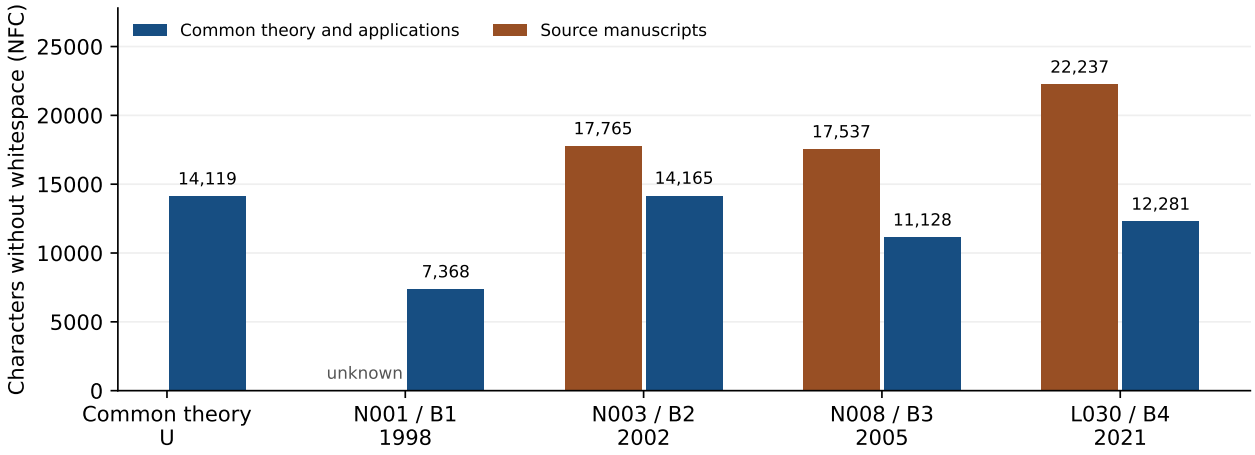
The three available source PDFs cost 57,539 characters. The original PDF for N001 is unavailable. If its unknown cost is x, the overall comparison is

$$R_B = \frac{x + 57{,}539}{59{,}061}.$$

There is no numerical value for R_B in this audit. B1 remains in the denominator; N001 is neither dropped nor assigned zero length.

Matched result	Source	Application	Conditional factor
N001 / B1	unknown	7,368	unknown
N003 / B2	17,765	14,165	1.254
N008 / B3	17,537	11,128	1.576
L030 / B4	22,237	12,281	1.811

Each conditional factor assumes that the *whole* common theory has already been acquired. Its cost is charged once in the complete-topic comparison above. Separate topics each pay their complete common theory, even where their arguments overlap. No finite mathematical program is required by this edition's uniform written proofs.



The blank source entry for N001 means unknown; it has no assigned bar height. The first stage is the common theory, followed by the four applications. The graph records actual document lengths and makes no claim of a general pattern of decreasing proof complexity.

What is counted

Every page is extracted with pypdf 6.13.0, in its original order. Titles, abstracts, mathematical statements and proofs, references, addresses, footnotes and unique publication information remain included. Only the individually recorded running headers and printed page numbers are removed. Printed hyphens remain; words are not joined across lines. The five new scientific PDFs comprise 22 pages. The three available source PDFs comprise 31 pages. Page counts are not used as the headline length unit. The two-page scope reader supplies audit context and no mathematical premise; it is identified separately.

The parser represents four pieces of large parentheses on N003 page 8 by private-use characters. Each is replaced by its visually identified Unicode hook, preserving four separate pieces. Four open diagram vertices on L030 page 5 are extracted as the letter i in a circle font; each becomes a circle character. Every occurrence is bound to its actual font, page offset and surrounding text. No other substitutions are made.

These rules were checked symmetrically across all eight available source and replacement PDFs. An independent program re-extracted all 53 pages and matched all 240 length cells and 48 normalized text hashes. It used independent counting code and rejected changes to the retained text boundary, page bindings, furniture and glyphs. The same PDF parser remains a shared dependency of the two counts.

Units and sensitivity

The full numerical ledger retains both NFC and NFKC normalization and three text variants: the complete raw extraction, that extraction after removing running furniture, and the selected text after glyph correction. For each it records five units: whitespace-separated tokens, maximal runs of letters and numbers, characters without whitespace, characters after collapsing whitespace, and UTF-8 bytes after the same collapse. An underscore separates such runs. Collapsing whitespace replaces each internal run by one space and removes leading and trailing whitespace. The ledger contains ninety exact known conditional fractions, together with explicit unknown entries for N001 and the complete topic. The headline coordinate above is fixed in advance of choosing a favorable factor; all other coordinates remain available in `results.json`.

Limits and attribution

These are lengths of the stated complete documents, not minimum proof complexities or counts of every visible mark. L030 page 5 has vector graphs whose edges and layout are not measured. Some graph labels extract despite being obscured by the drawing. N008's bibliography underline strokes extract as minus glyphs; they remain counted. N003's original PDF contains repeated font dictionary keys; the source bytes and parser are fixed, and its page and font extraction are bound rather than silently repaired.

The foundational Buekenhout–Cara–Dehon classification and the two separate Whiston results remain substantial retained inputs. Cayley's formula, the stated symmetric-group automorphism fact, the finite p-group Frattini theorem, Zorn's lemma and elementary linear algebra also remain explicit. Their expanded proof costs are unknown. Lean compilation and finite hypothesis counterexamples are verification infrastructure; they do not supply an omitted mathematical enumeration. No complete source application is claimed to be formalized in Lean.

This audit preserves the existing source and question credits. The mathematical admission and corrected source map specify the thirty clauses being compared; this reader does not expand that scope.